DQLab

live Class

Social Media Analytics



Bootcamp Data Analyst with SQL & Python using Google Platform

Fadia Nur Adiba

PROJECT OVERVIEW

OBJECTIVE

To uncover how user behavior and platform performance influence engagement and social commerce outcomes, and to provide actionable insights for data-driven marketing decisions.

Data Period: Jan 2022 - Dec 2024

KEY ANALYSIS AREAS

- User Demographics & Activity
- Engagement & Behavior Pattern
- Platform Performance
- Content Performance
- Social Commerce Performance

TOOLS & DATASET

TOOLS

ETL & Preparation
Google Bigquery

EDA
Google Colab

Visualization
Data Studio

Persentation
Canva

DATASET

Source
Social Media User Behavior (Kaggle)

Dataset Size
2,000 Users

Business Focus
Engagement, Conversion & Monetization

Exploratory Data Analysis (EDA)

The initial stage to understand the data and identify key patterns that will be analyzed further

01 What is the overall user engagement performance?

02 Which platform has the highest engagement?

03 What age group dominates social media activity?

04 Which countries contribute the highest number of users?

05 During which time period are users most active?

06 How does engagement/purchase behavior change over time?

07 Is user engagement/purchase behavior stable or seasonal across years?

Exploratory Data Analysis (EDA)

Based on Overview Dashboard

2,000

Users

Solid user base

#1

Instagram

Highest engagement

3 Hours

Avg. daily usage

Strong activity

33.9

Engagement

Active interaction

61%

Avg. Purchase Rate

High conversion

$22.6

Avg. Monthly Spend

Monetization potential

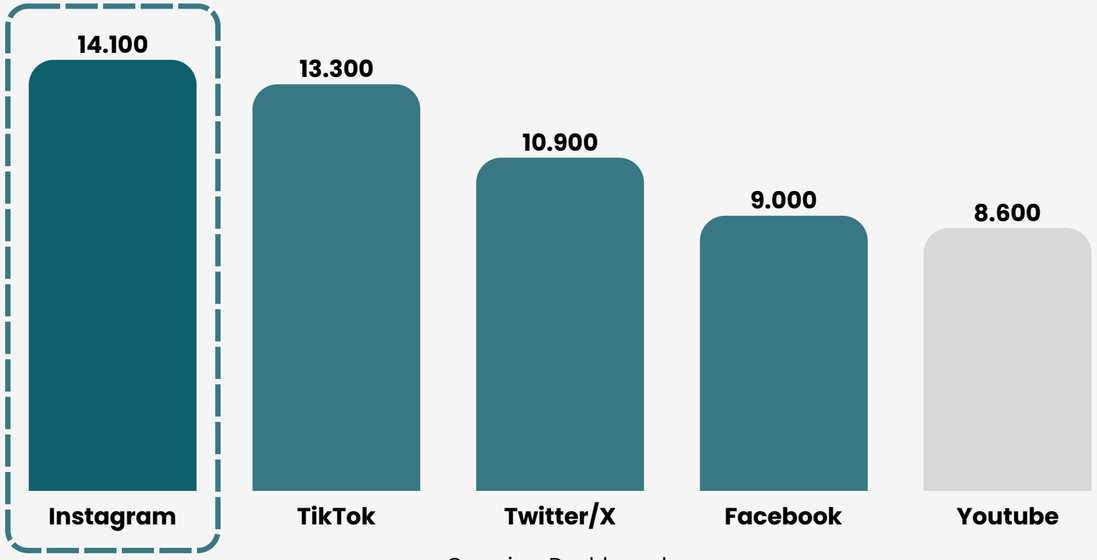
INSIGHT

- Avg engagement of 33.9 and 3-hour usage indicate highly active user interaction patterns.
- 61% purchase rate reflects strong conversion despite moderate spending behavior.
- $22.6 avg. monthly spend indicates solid monetization opportunity.

Exploratory Data Analysis (EDA)

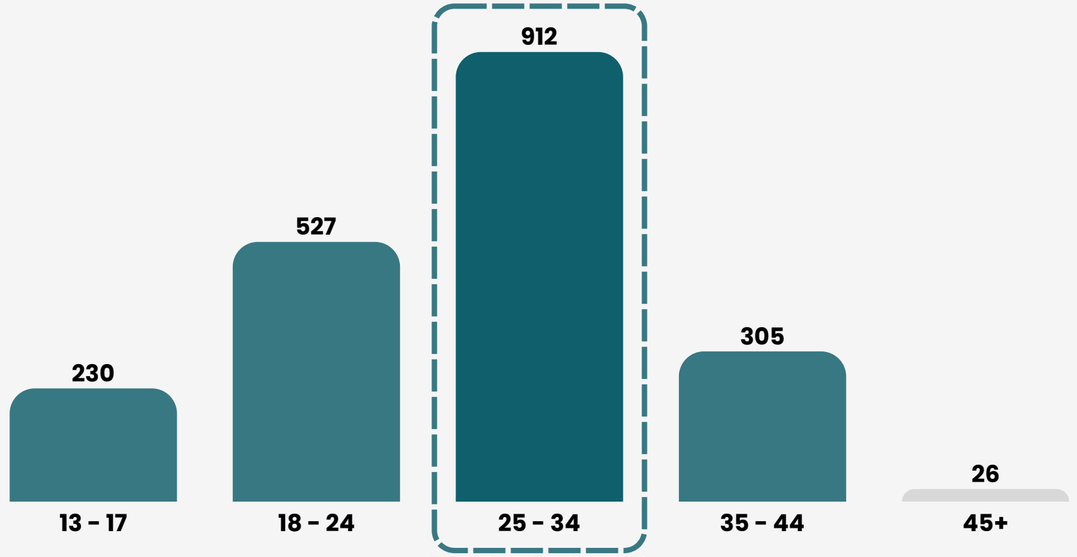
Based on Overview Dashboard

Top Engagement by Platform



Overview Dashboard

Age Group Distribution



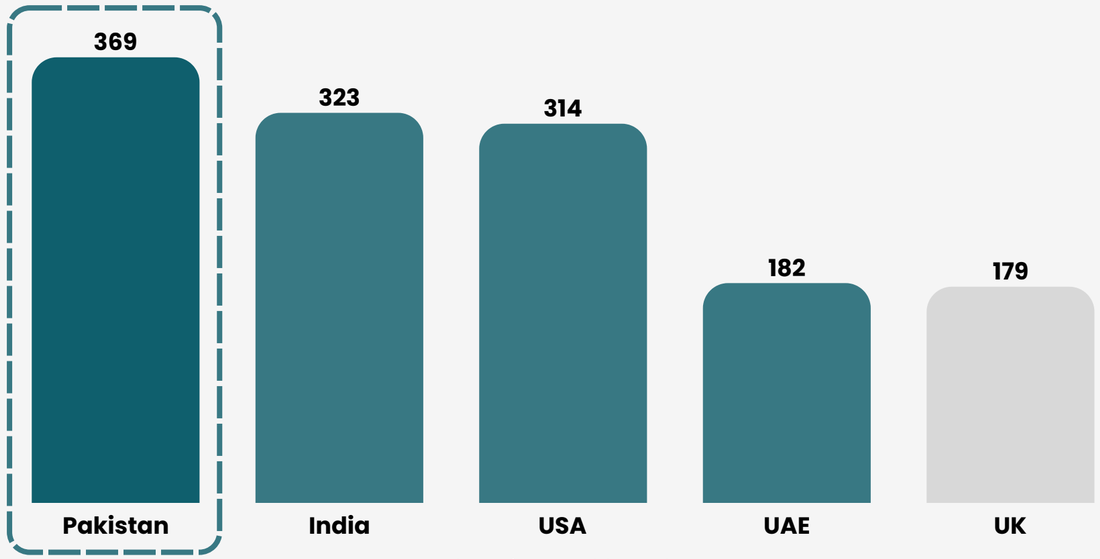
Overview Dashboard

Instagram (14.1K) and TikTok (13.5K) generate **the highest engagement,** indicating that users aged **18–34** prefer **fast, visual, and interactive content formats** that support **high-frequency social interaction.**

Exploratory Data Analysis (EDA)

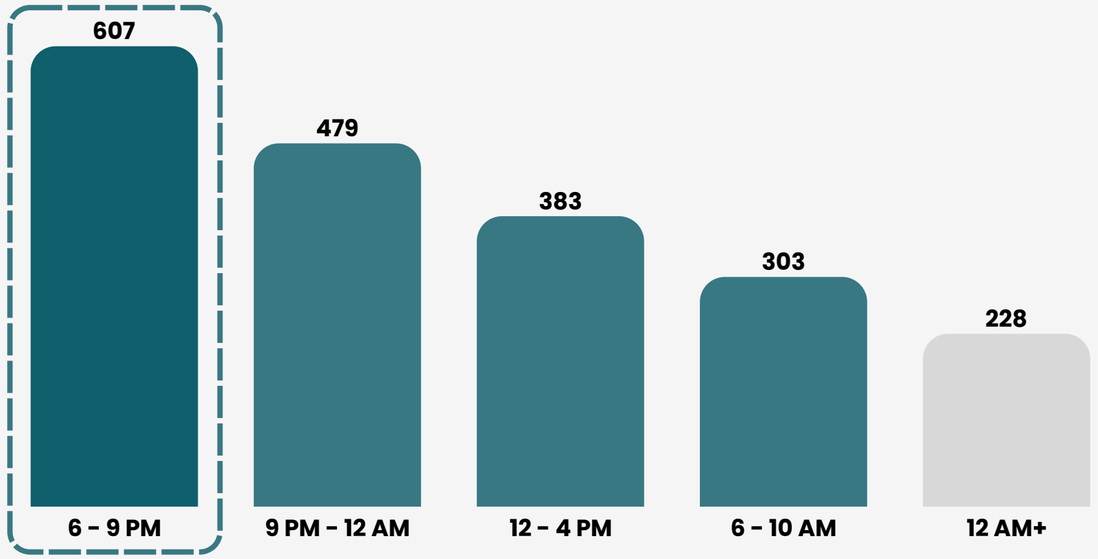
Based on Overview Dashboard

Top Country by Users



Overview Dashboard

Peak Usage Time

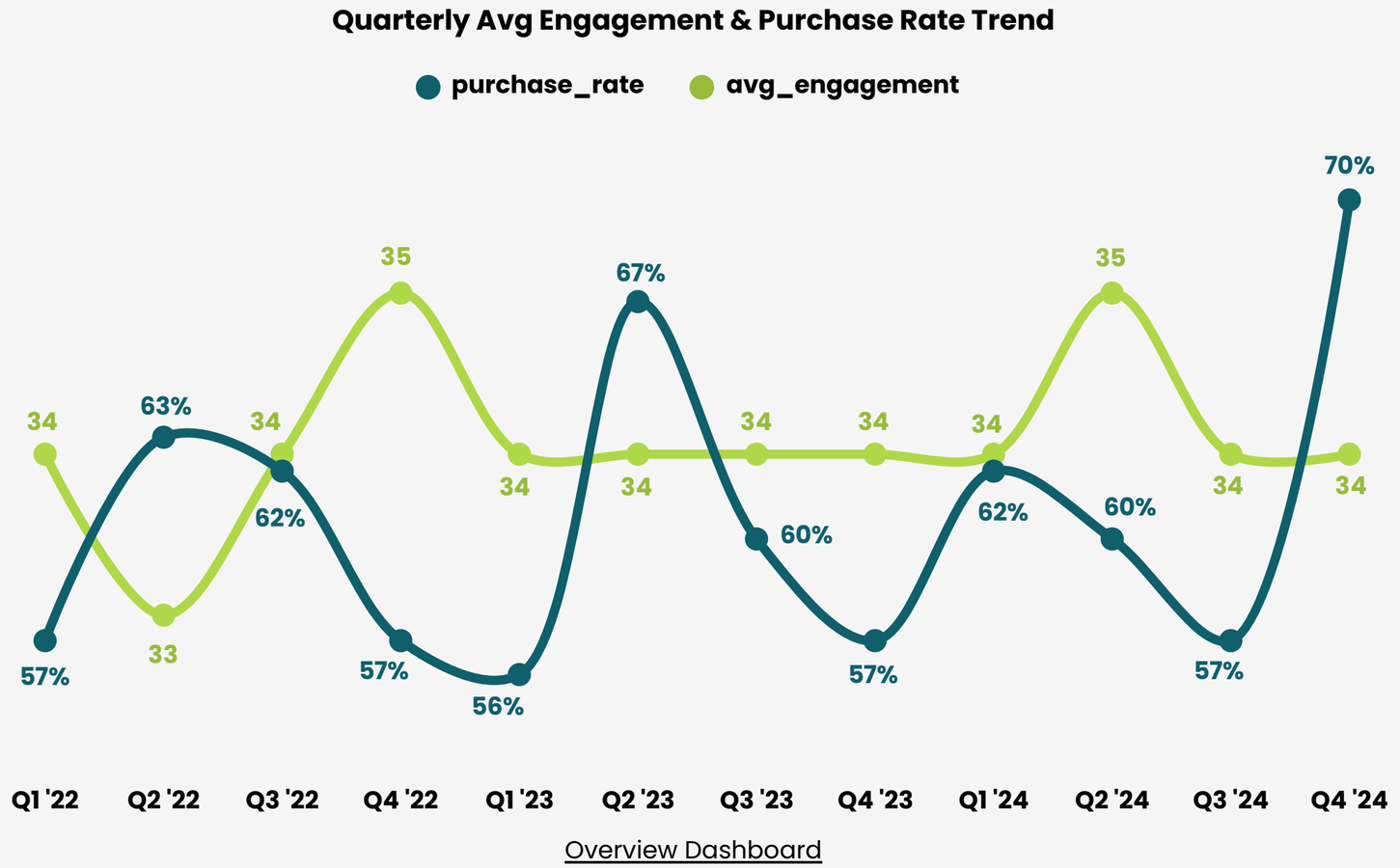


Overview Dashboard

Users from **Pakistan** contribute **the highest number of users**, while **peak activity** occurs during **the Evening period,** indicating strong engagement from users **who are most active after working or study hours**.

Exploratory Data Analysis (EDA)

Based on Overview Dashboard



Overview Dashboard

INSIGHT

Engagement remain stable across quarters.

Avg engagement stays consistent around 34%-35% throughout the period

Purchase rate fluctuates across quarters.

Purchase rate varies between 56% and 70% showing no consistent pattern.

Higher engagement ≠ Higher purchase rate

High engagement levels do not directly drive conversion performance.

Deep Dive Questions

Analytical questions used to uncover deeper insights, patterns, and user behavior across multiple dashboards

01 How do different user segments influence engagement and usage behavior?

02 How do user activity patterns affect engagement throughout the day?

03 Which social media platform demonstrates the strongest engagement performance?

04 What type of content and interaction contributes to higher engagement across platforms?

05 How do user segments and platforms influence purchase behavior and revenue generation?

Deep Dive Analysis

Based on User Segmentation Dashboard

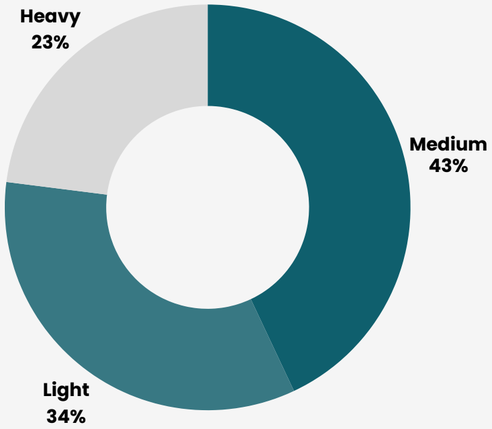
43%
Medium Segment
Users
Largest user base

44%
Medium Segment
Highest Engagement
Drive by largest user base

44%
Heavy Segment
Highest Daily Usage
Highest Usage Intensity

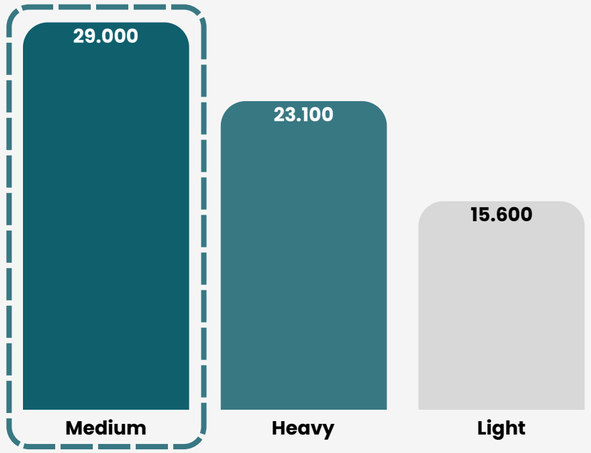
68%+
Smartphone
Mobile-first behavior
Primary Device

User Segment Distribution



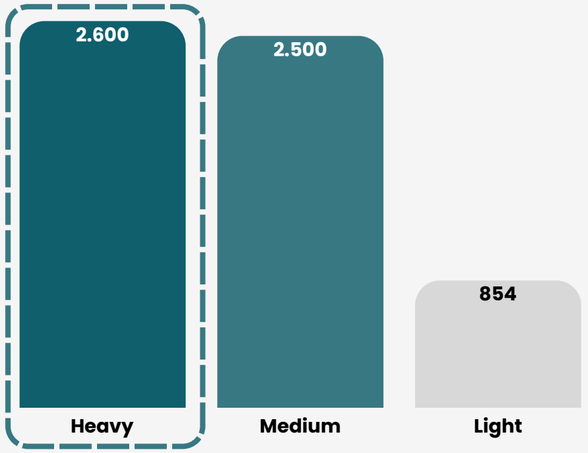
User Segmentation Dashboard

Engagement by Segment



User Segmentation Dashboard

Daily Usage (Hours) by Segment



User Segmentation Dashboard

INSIGHT

Consistent moderate activity is more scalable than extreme usage behavior,

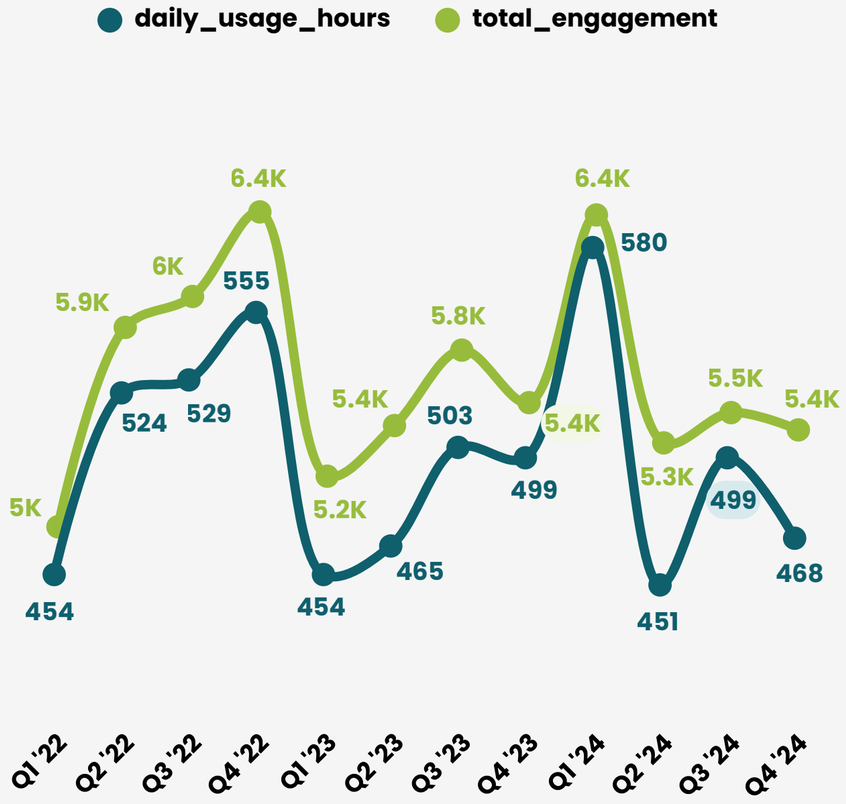
Medium → Highest engagement
Heavy → Highest usage intensity
Light → Lower activity levels.

Daily Usage → Heavy > 4 hours Medium 2 - 4 hours Light < 2 hours

Deep Dive Analysis

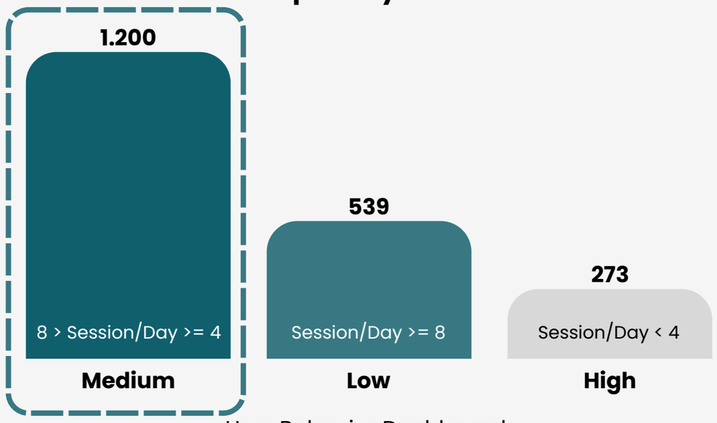
Based on User Behavior Dashboard

Daily Usage (Hours) & Engagement Trend



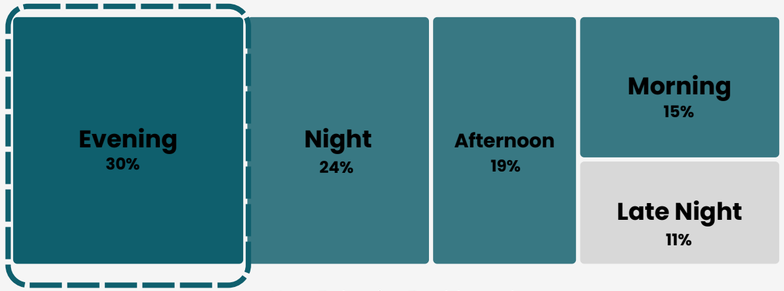
User Behavior Dashboard

Sessions per Day Distribution



User Behavior Dashboard

Active Usage by Time Period



User Behavior Dashboard

INSIGHT

Engagement follows usage trend.

Higher daily usage tends to generate higher engagement levels.

Medium session users dominate activity.

Users with 8+ sessions/day contribute the highest activity shares.

Evening is the most active period.

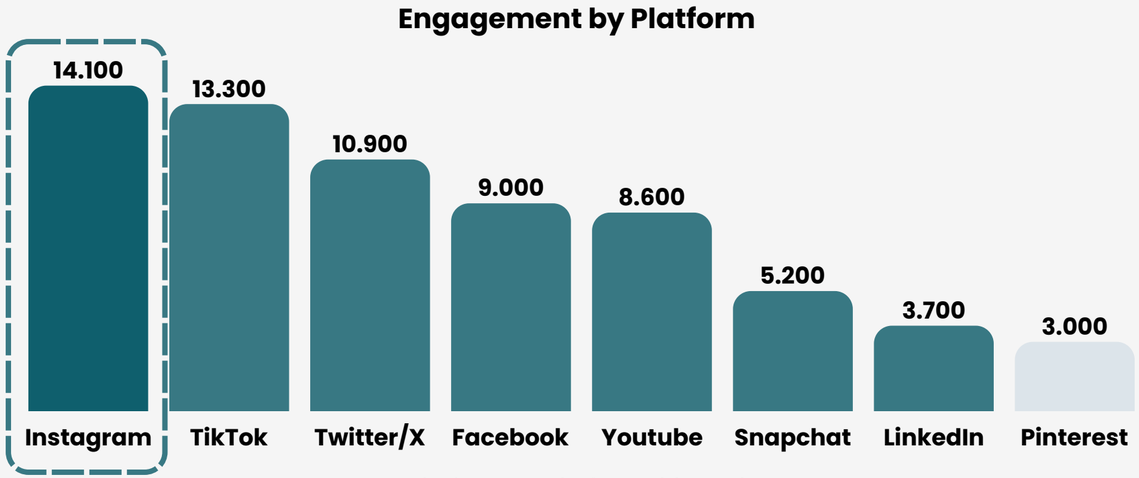
Most users are active during evening hours.

Consistent behavior pattern observed.

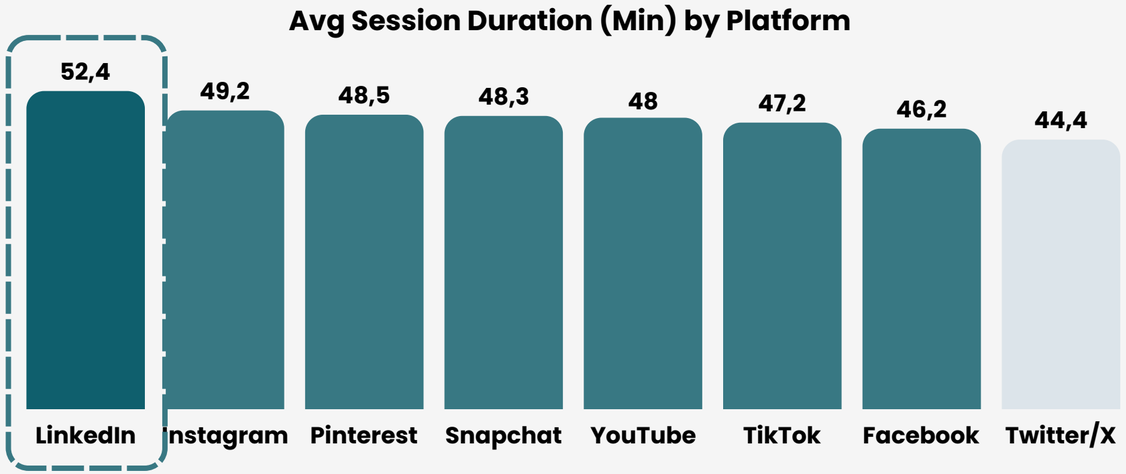
Usage and engagement move in similar patterns across quarters.

Deep Dive Analysis

Based on Platform Analysis Dashboard



Platform Analysis Dashboard



Platform Analysis Dashboard

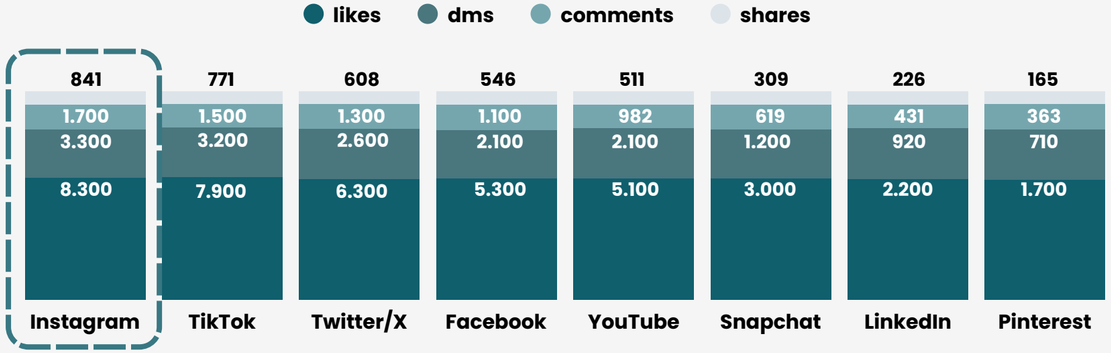
INSIGHT

- Instagram contributes the highest engagement at 21%, showing the most user interaction
- LinkedIn records the longest average session duration (52.2 min), indicating higher user retention.
- Longer sessions ≠ higher engagement, as Instagram drives more active interactions.
- TikTok also maintains strong engagement performance across both metrics.

Deep Dive Analysis

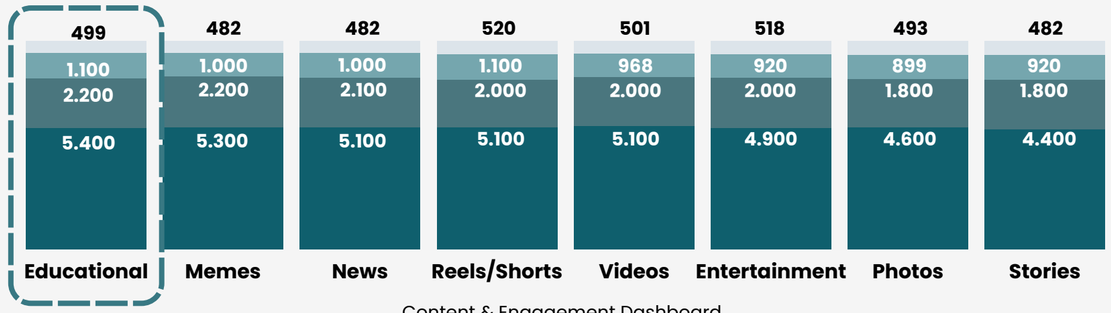
Based on Content & Engagement Dashboard

Engagement by Platform



Content & Engagement Dashboard

Engagement by Content Type



Content & Engagement Dashboard

Content vs Platform Engagement

	Instagram	TikTok	Twitter/X	Facebook	YouTube	Snapchat	LinkedIn	Pinterest
Educational	1.8K	1.6K	1.4K	1.2K	1.6K	371	724	576
Memes	1.9K	1.9K	1.6K	1.1K	891	495	646	378
News	2K	1.7K	1.3K	1.1K	1.1K	788	471	344
Reels/Shorts	1.8K	1.5K	1.3K	1.5K	829	798	597	293
Videos	1.6K	1.9K	1.4K	1.1K	1.1K	635	397	422
Entertainment	1.6K	1.6K	1.8K	961	1K	677	332	283
Photos	1.5K	1.9K	1.1K	1.1K	937	714	249	385
Stories	1.9K	1.3K	1.1K	976	1.1K	695	332	300

INSIGHT

- Instagram and TikTok → strongest platform engagement.
- Educational content drives broad engagement, while Instagram and TikTok respond more strongly to entertaining and fast-consumption content.
- Likes and comments → dominant interaction drivers, reflecting fast and active engagement behavior.

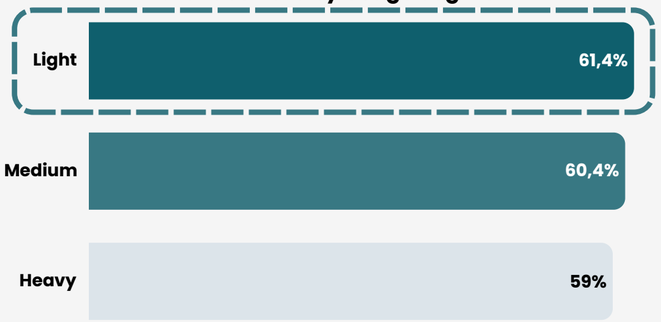
Deep Dive Analysis

Based on Social Commerce Dashboard

Top Converting Platform

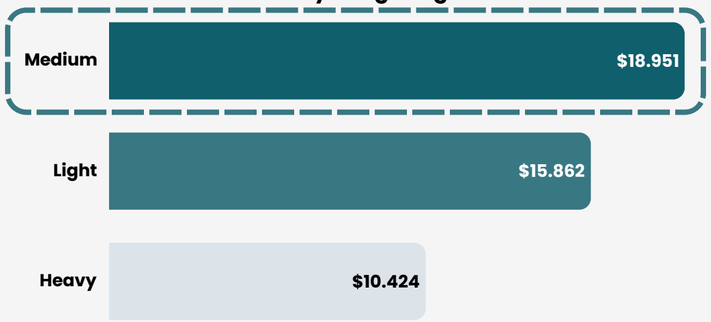
Platform	Purchase Rate	Spend per Purchased	Revenue
Tiktok	64%	$39.3	$10.0K
Instagram	59%	$37.4	$9.2K
Twitter/X	57%	$40.6	$7.3K
Facebook	63%	$35.7	$6.1K
YouTube	62%	$34.5	$5.3K
Snapchat	61%	$36	$3.5K
LinkedIn	54%	$36.2	$2.1K
Pinterest	57%	$34.4	$1.7K

Purchase Rate by Usage Segment



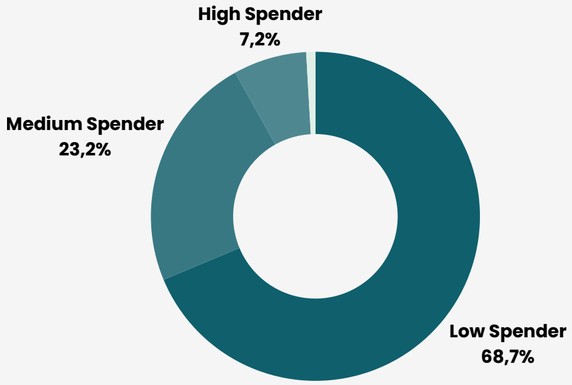
Social Commerce Dashboard

Revenue by Usage Segment



Social Commerce Dashboard

Spender Segment Distribution



Social Commerce Dashboard

INSIGHT

- Low spenders → dominant user base
- Light usage → highest purchase rate.
- Medium usage → highest revenue.
- TikTok → highest revenue & purchase rate
- Twitter/X → highest spend.

Monthly Spend → Premium > $150, High $75 - $150, Medium $25 - $74, Low < $25

KEY INSIGHTS

01



Consistent Moderate Activity Drives Engagement

Medium users contribute the highest engagement volume, supported by dominant medium session frequency (4-8 sessions/day)

02



Users Prefer Fast & Interactive Experience

Strong Instagram and TikTok engagement, smartphone dominance, and evening activity indicate a preference for visual and interactive content.

03



Daily Usage & Engagement Move Closely Together

Daily usage and engagement trends move closely together, indicating that higher user activity generally leads to stronger engagement performance.

04



Engagement Does Not Guarantee Conversion

Engagement remains stable across time, while purchase rate fluctuates, indicating conversion is influenced by factors beyond engagement.

05



Strong Conversion, Moderate Monetization

A 61% purchase rate reflects strong user participation in social commerce activities, but the dominance of low spenders (68.7%) limits revenue growth potential.

06



Engagement Drivers

Instagram & TikTok



Conversion Drivers

TikTok



High-Value Spending Driver

Twitter/X

SUMMARY

Social media performance is primarily driven by **consistent user activity**, **platform preference**, and **content strategy**, while monetization growth depends on improving **conversion value** rather than increasing particioation alone.

AUDIENCE

- Medium users dominate engagement.
- Peak activity during evening hours.
- Smartphone-first behavior.

PLATFORM

- Instagram & TikTok lead engagement
- LinkedIn drives longer sessions.
- Different platforms serve different user behavior.

CONTENT

- Educational, news, memes, and videos perform best.
- Users prefer fast, practical, and visual content.
- Short-form content drives stronger engagement.

COMMERCE

- TikTok leads conversion and revenue.
- Twitter/X generates the highest spending.
- Monetization opportunity remains underutilized

RECOMMENDATIONS

01



PRIORITIZE INSTAGRAM & TIKTOK

Allocate more marketing resources to platforms with the highest engagement and conversion performance.

02



OPTIMIZE POSTING TIME

Schedule content during evening - night hours (6 PM - 12 AM) to maximize reach and user interaction.

03



INCREASE SHORT-FORM & EDUCATIONAL CONTENT

Expand the use of Reels, TikTok videos, and educational content

04



STRENGTHEN MONETIZATION STRATEGY

Increase average spending through upselling, product bundles, and premium offerings.

05



TARGET HIGH-VALUE SEGMENTS

Develop retention and loyalty strategies for Medium and Heavy Users

06



IMPROVE COMMERCE STRATEGY

Use TikTok for conversion-focused campaigns while leveraging Instagram for engagement and Twitter/X for high-value spending audiences.

07



REDUCE LOW-PERFORMING CONTENT

Minimize static photo-based content and shift toward more interactive and dynamic formats

Thank you, hope this analysis provides valuable insights!

Fadia Nur Adiba

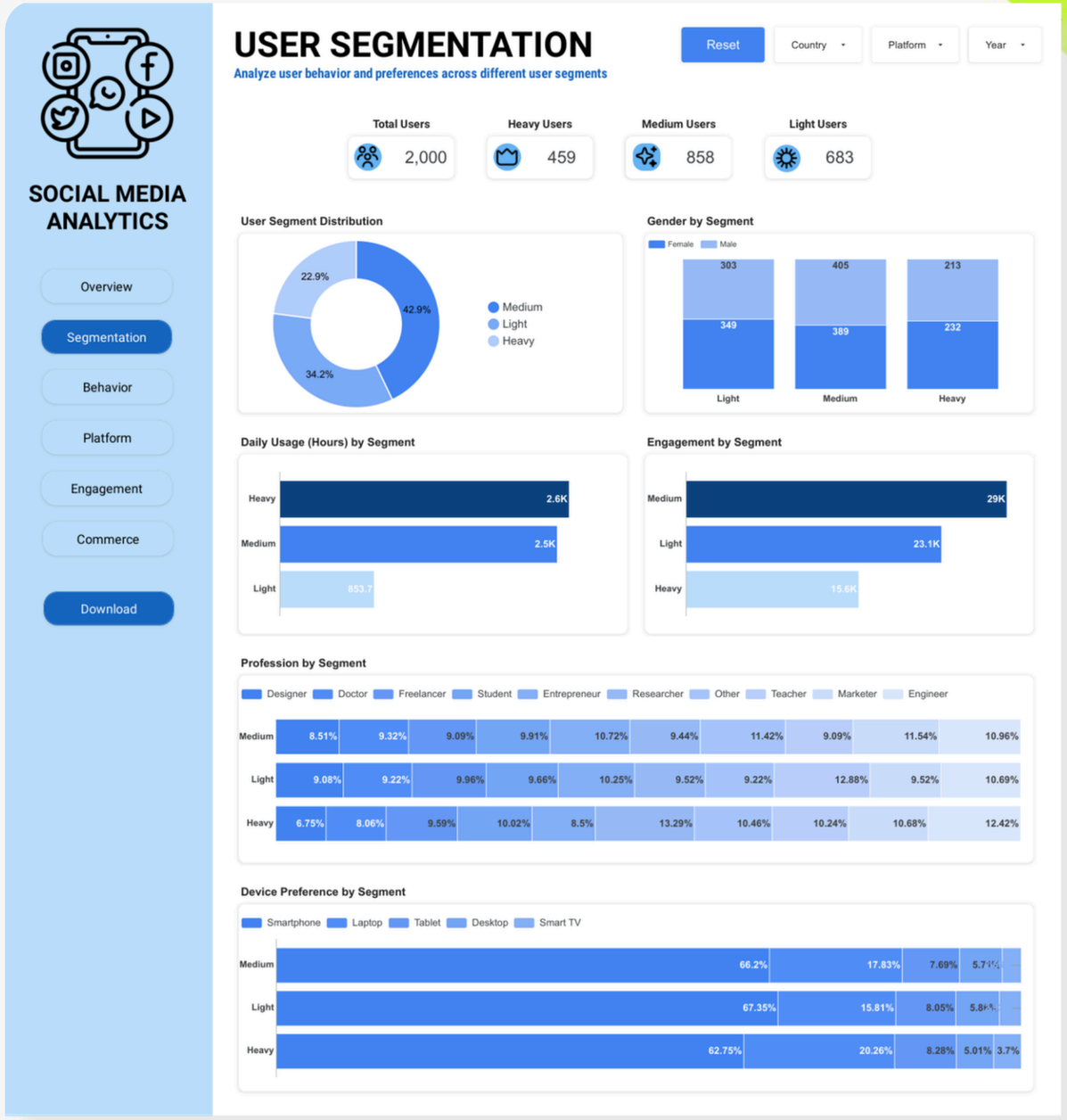
nuradibafadia@gmail.com

(+62) 82121260026

Overview Dashboard



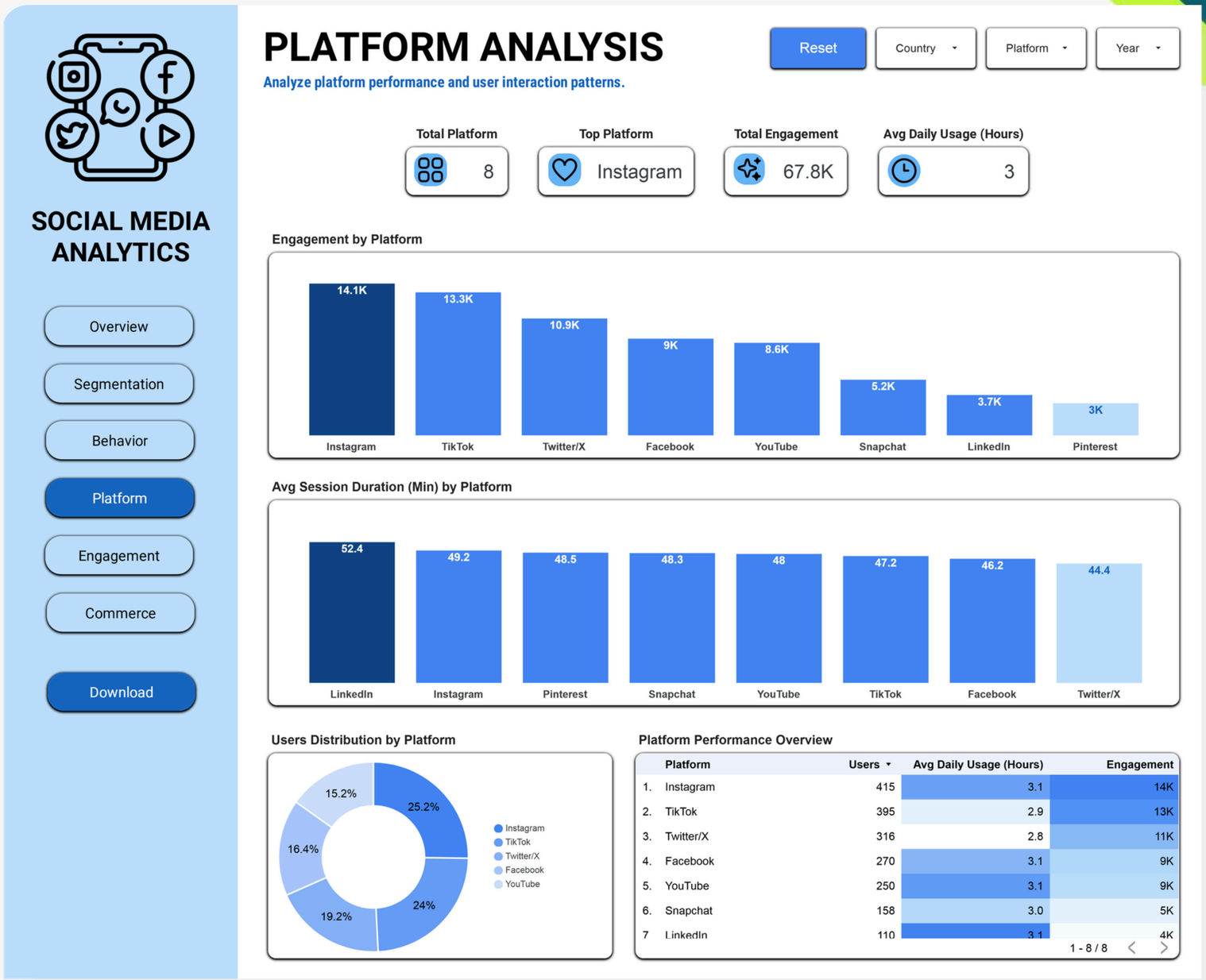
User Segmentation Dashboard



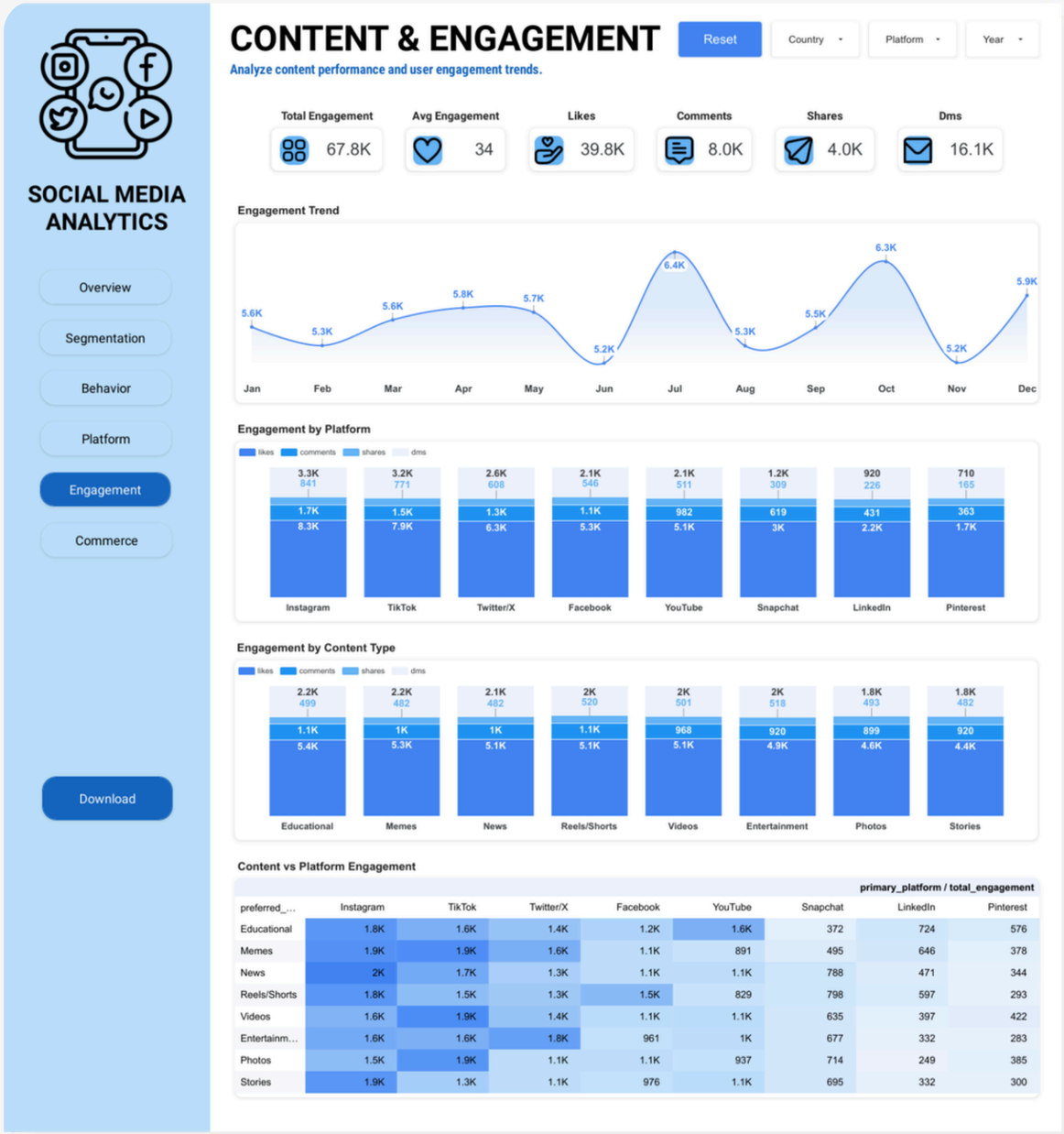
User Behavior Dashboard



Platform Analysis Dashboard



Content & Engagement Dashboard



Social Commerce Dashboard

